ACE Cyber Project Research Design

1. Project Information

Field	Response
Student Name(s)	Denzel Jefferson
University	Charleston Southern University
Faculty Advisor	Professor Julie Henderson
DoD/ACE Mentor	Professor William Littleton
Project Title	Contextualized Threat Intelligence Prioritization for CTI Analysts
Version	v3.0
Date	May 4, 2026

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2. Research Method Selection

Select one:

■ Experimental ■ Applied ☐ Descriptive ☐ Exploratory

☐ Mixed Methods

Briefly describe why this method was chosen:

(Explain how this method best aligns with your research question and objectives.)

This research study requires an applied, experimental approach because it focuses on developing and evaluating an automated threat intelligence prioritization ranking system. This research will explore how operational environment data from Elasticsearch can be integrated into a weighted contextual scoring model to enhance CTI analyst efficiency and ultimately reduce alert fatigue.

3. Research Design Overview

Provide a high-level description of your research design: This research design will examine how operational environment data collected from Elasticsearch can be used to automate and contextualize threat intelligence prioritization within a contextualized threat prioritization framework.

- What are you studying?

- What are the key variables or factors?

    1. Threat prioritization accuracy
    2. CTI analyst efficiency and workload
    3. Automated impact on alert fatigue
    4. Volume of alerts refined

- What is your overall approach (quantitative, qualitative, or mixed)? My overall approach is primarily quantitative with some qualitative aspects.
- What do you expect to learn or demonstrate? I want to demonstrate that contextual operational data can significantly improve automated threat prioritization workflows, enabling faster CTI analyst decisions and reducing alert fatigue.

Example: "This study will use a controlled experiment to test how a conversational AI model improves log query efficiency compared to baseline manual methods."

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4. Research Objectives & Hypothesis

Primary Objective(s):

1. Develop a platform that integrates Elasticsearch operational data into a contextualized threat-prioritization model.
2. Analyze the automated results and determine their impact on CTI analysts' efficiency.
3. Evaluate the reduction of alert fatigue through contextualized threat prioritization scores.

Hypothesis (if applicable):

If operational environment data from Elasticsearch is incorporated into an automated MISP-based (simulated) threat-prioritization system, then the time required for CTI analysts to identify high-priority threats will significantly decrease while maintaining or improving overall prioritization accuracy compared to a manual review process.

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5. Planned Research Steps / Experimentation Plan

Outline your step-by-step approach to conducting your research.
Each step should be clear and actionable.

Step	Description	Expected Output	Duration / Timing
1	Define and refine research question	Finalized research focus	Week 1

2	Conduct literature review	Literature map	Week 2
3	Develop prototype / model / framework	Initial build	Week 4
4	Generate synthetic log datasets and and execute prioritization tests	Testing datasets	Week 5

Step	Description	Expected Output	Duration / Timing
5	Execute experimental test cases	Collected performance data	Week 6
6	Analyze results	Findings summary	Week 8
7	Prepare report & presentation	Final deliverables	Week 10

Adjust as appropriate for your research timeline.

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6. Data Collection Plan

Describe how you will collect your data, what type of data you expect (quantitative, qualitative, or both), and how you will ensure accuracy and reliability.

Data Source	Type (Quant / Qual)	Description	Collection Method	Tool / Instrument	Notes
Elasticsearch Logs (Log Generation)	Quantitative	Operational event data	Automated log generation/ consumption	Elasticsearch	Simulated environment and synthetic logs
Contextual scoring model output	Quantitative	Threat prioritization scoring output	Automated logs	Internal scoring logic (rule-based model)	Compare scores before and after
Analyst workflow metrics	Quantitative	Analyzing metrics	Examining workflow metrics	Python script	Measure efficiency

7. Data Analysis Plan

Explain how you will analyze your data to test your hypothesis or address your research question.

- What statistical tests, visualizations, or coding methods will you use?

- How will you interpret qualitative responses (e.g., coding themes, content analysis)?
- How will success be measured (KPIs, accuracy, latency, efficiency, satisfaction, etc.)?

Quantitative data will be collected from Elasticsearch logs and prioritization outputs from the contextual scoring model to analyze and evaluate differences between contextualized threat rankings and basic rankings. Qualitative analysis is minimal, as the study does not include human participants; thus, the metrics below will serve as the measure of success. Descriptive statistics will be used to measure success in reducing alert volume, improving prioritization accuracy, and increasing analyst efficiency.

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8. Research Tools, Platforms, and Resources

List all tools, environments, or resources you will use during the project.

Tool / Resource	Purpose	Access Needed?	Notes
Elasticsearch Enterprise	Data analysis	Yes	Provided by ACE Lab
Python (Jupyter Notebook)	Experiment scripting	No	Installed locally
MS Excel / SPSS	Data analysis	No	

9. Risks & Mitigations

Identify potential risks or challenges that may affect the success or validity of your study, and describe how you plan to mitigate them.

Risk	Potential Impact	Mitigation Strategy
Limited data diversity	May bias results	Use multiple sample sources

Tool access delay	Timeline impact	Request early access
Synthetic logs	May reduce the effectiveness of results	Use realistic log entries and evaluate against varied synthetic datasets

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10. Ethical & Data Handling Considerations

- Will your study involve human participants (e.g., surveys, interviews)?
- Will you handle sensitive or anonymized data?
- How will you ensure compliance with ethical guidelines, FERPA, or DoD data protection rules?

This study will not involve human participants, such as surveys or interviews. Instead, all data used will be simulated, derived from operational logs, or derived from threat intelligence data.

All research will follow university guidelines and DoD data protection policies. All datasets used will be confined to non-sensitive test environments to guarantee ethical research practices.

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11. Expected Outputs

List what deliverables or measurable outputs will result from your research process.

Deliverable	Description	Target Date
Prototype	Functional contextualized threat intelligence prioritization prototype	Week 6
Data Analysis Report	Summary of results	Week 8
Final Presentation	15-20 minutes	Week 10

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12. Mentor Feedback / Sign-Off

Reviewer	Signature / Initials	Date	Comments
Faculty Advisor			

DoD/ACE Mentor			



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